Robustness Audit of Health Misinformation Classification Models: Progress Report

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Introduction

Health misinformation detection is commonly framed as binary text classification: a model receives health-related content and predicts whether it is reliable or unreliable. That framing is useful, but incomplete. Model performance can reflect genuine textual signals, but it can also reflect publisher identity, topic concentration, COVID-era vocabulary, or other dataset artifacts. The study therefore evaluates whether classifiers for dataset-defined unreliable health information learn robust signals or mainly exploit shortcuts tied to specific datasets.

CoAID is used as the primary COVID-19 health-information dataset, and FakeHealth is used as the secondary broader health-news dataset. The outcome is described as "dataset-defined unreliable health information" rather than verified claim-level misinformation because the two datasets construct labels differently. CoAID is organized around real and fake COVID-19 news and claims, whereas FakeHealth is based on HealthNewsReview ratings and review criteria. Treating those labels as interchangeable truth labels would overstate what the datasets can support.

Background and Hypotheses

The reviewed literature supports a robustness-audit design. Systematic reviews show that health misinformation is a public-health problem across topics and platforms, while also emphasizing variation in definitions, data sources, and measurement strategies (Suarez-Lledo & Alvarez-Galvez, 2021; Wang et al., 2019). That variation matters because a model trained to reproduce expert ratings, source-level credibility, or fact-check labels may perform well while answering different scientific questions.

Prior technical work also motivates controlled evaluation. Baris-Schlicht et al. (2023) identify recurring limitations in automatic health misinformation detection, including dataset constraints, topic differences, and limited cross-dataset testing. Di Sotto and Viviani (2022) argue that detection may involve textual, emotional, medical, propagation, and user-context features, and they discuss both CoAID and FakeHealth as datasets with different units and metadata. FakeHealth is especially useful because it links article text to expert health-news reviews and social engagement identifiers (Dai et al., 2020), but its engagement metadata must be evaluated for coverage before it can be used as a predictor.

The primary hypothesis is that traditional text models will produce strong within-dataset performance, especially on CoAID, but that performance alone will not demonstrate generalizable misinformation understanding. A secondary hypothesis is that social engagement features will add value only where metadata coverage, class balance, and proxy-risk checks justify their use. A third hypothesis is that transformer models may improve text classification, but only if evaluated against the same frozen splits as the traditional baselines.

Materials and Methods

The implemented methodology is a reproducible robustness-audit pipeline. Raw datasets are preserved under `data/raw/` and excluded from GitHub. Acquisition code documents the dataset sources, including the optional FakeHealth Zenodo user-network archive, which has been downloaded locally. Inventory, provenance, and label scripts then document source files, available fields, native labels, harmonized labels, inclusion rules, and compatibility constraints.

Processed tables standardize CoAID and FakeHealth into a common schema. CoAID article and claim rows are classification units. FakeHealth HealthStory and HealthRelease review/content records are joined by `news_id`. The shared schema contains identifiers, text fields, native and harmonized labels, source metadata, engagement counts, optional user-network availability, and exclusion reasons. FakeHealth review explanations are excluded from model inputs so that outcome-bearing review text is not used as a predictor.

Evaluation uses frozen split manifests rather than ad hoc resampling. The standard evaluation applies stratified duplicate-group splitting. The controlled evaluation attempts source-disjoint splitting while preserving duplicate-family separation; if that constraint cannot be satisfied, the pipeline records the reason and falls back to duplicate-group stratified evaluation. Traditional base-

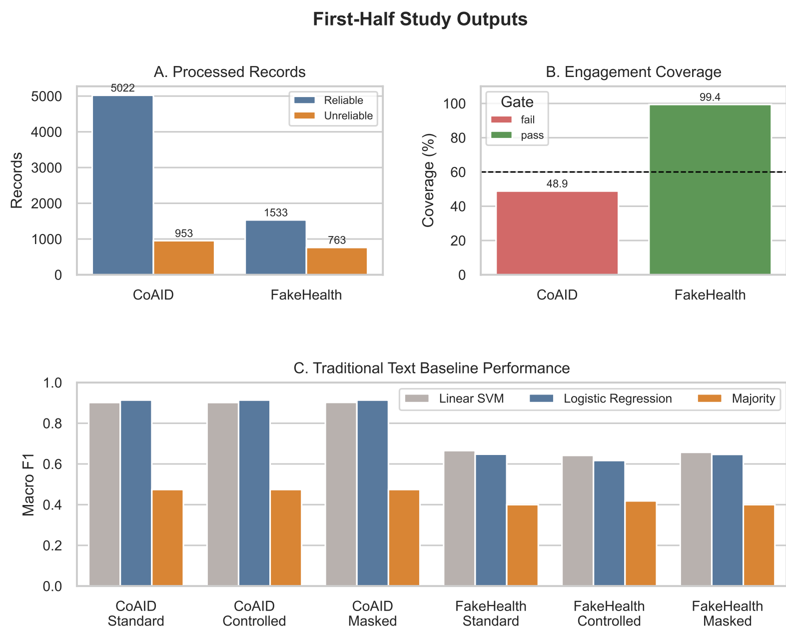

Figure 1: First-half study outputs: processed records, engagement feasibility, and traditional text baseline performance.

lines include a majority classifier, TF–IDF Logistic Regression, and TF–IDF Linear SVM. Metrics include macro F1, class-specific precision/recall/F1, PR–AUC for the unreliable class, confusion matrices, and bootstrap intervals.

Results to Date

The first-half implementation has produced local outputs for inventory, provenance, profiling, splitting, and traditional model evaluation. The processed dataset contains 8,271 included records: 5,975 from CoAID and 2,296 from FakeHealth. As shown in Figure 1, both datasets are class-imbalanced but contain enough reliable and unreliable records for within-dataset text modeling.

The engagement feasibility check produced a clear boundary for later analysis. CoAID failed the predictive engagement gate because usable engagement metadata covered only 48.9% of included records and the unreliable class had only 166 engagement-complete records. FakeHealth passed the gate with 99.4% engagement coverage, 1,526 reliable engagement-complete records, and 756 unreliable engagement-complete records. This does not yet show whether engagement improves prediction, but it prevents an underpowered or biased engagement experiment on CoAID.

The split audit also produced an important methodological result. FakeHealth passed source-disjoint splitting. CoAID did not: the first source-disjoint attempt violated duplicate-family separation, so the pipeline fell back to duplicate-group stratified evaluation. The final split manifests guarantee that no duplicate group crosses train, validation, and test partitions.

Traditional text baselines have been run for CoAID and FakeHealth across standard, controlled, and masked-text conditions. CoAID results are strong: Logistic Regression reaches approximately 0.914 macro F1 under standard, controlled, and masked conditions, while Linear SVM reaches approximately 0.902. FakeHealth is more difficult: the best standard result is Linear SVM at approximately 0.666 macro F1, and controlled FakeHealth performance is lower. These early results support the hypothesis that within-dataset performance is dataset-dependent.

Discussion and Next Steps

The first research question is partially answered. Traditional text baselines are complete, but the transformer comparison is not complete because the local environment lacks `torch`, `transformers`, `datasets`, and `evaluate`. The next technical step is to install those transformer dependencies and rerun the DistilBERT evaluation script. The next scientific step is to compare DistilBERT against the frozen traditional baselines before evaluating engagement ablations, cross-dataset transfer, narrative structure, calibration, selective prediction, and explanation auditing.

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